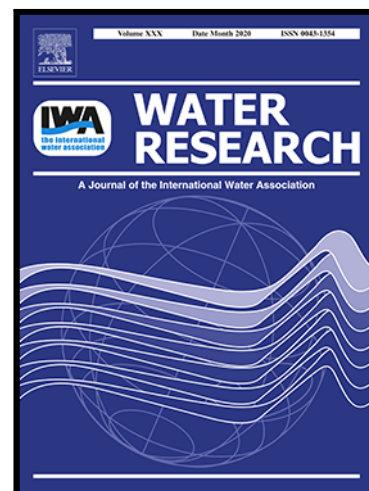


Antibiotics and antibiotic resistant genes (ARGs) in groundwater: A global review on dissemination, sources, interactions, environmental and human health risks

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PII: S0043-1354(20)30990-8
DOI: <https://doi.org/10.1016/j.watres.2020.116455>
Reference: WR 116455



To appear in: *Water Research*

Received date: 11 July 2020
Revised date: 22 September 2020
Accepted date: 23 September 2020

Please cite this article as: Syeda Maria Zainab , Muhammad Junaid , Nan Xu ,
Riffat Naseem Malik , Antibiotics and antibiotic resistant genes (ARGs) in groundwater: A global
review on dissemination, sources, interactions, environmental and human health risks, *Water Re-
search* (2020), doi: <https://doi.org/10.1016/j.watres.2020.116455>

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HIGHLIGHTS

- Global groundwater occurrence of antibiotics and antibiotic resistance genes reviewed
- Quinolones, sulfonamides, tetracycline and their genes most prevalent in groundwater
- Targeted groundwater antibiotics not proven to pose human health risks
- Significant correlations exhibit among antibiotics, bacterial communities, environmental factors, and ARGs
- Groundwater is an important environment to investigate the spread and transport of antibiotics and ARGs

Review article

Antibiotics and antibiotic resistant genes (ARGs) in groundwater: A global review on dissemination, sources, interactions, environmental and human health risks

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Abstract

The discovery and evolution of antibiotics for humans and animals are among the most significant milestones of the 20th century. However, antibiotics play a significant role in the induction and dissemination of antibiotic resistance genes (ARGs) in groundwater that has recently become the primary environmental concern. They are administered to humans and animals on a large scale and are persistent in the environment. Long term impacts of antibiotics in the ecological environment are not still clearly understood, and their occurrence and consequences have become an important research topic worldwide. The hotspot reservoirs of antibiotics and ARGs include medical facilities, livestock farming, aquaculture, landfills, on-site sanitation systems, sewage, and wastewater treatment plants. Our meta-analysis demonstrated that antibiotics, including ciprofloxacin, sulfamethoxazole, erythromycin, and tetracycline were found at high concentrations while sulfonamide and tetracycline ARGs were more prevalent in groundwater. Moreover, the highest reported concentrations of targeted antibiotics were used to calculate hazard quotient (HQ) and risk quotient (RQ) in global groundwater bodies to estimate environmental and human health risks, respectively. Due to limited available ecotoxicity data, RQ and HQ can only be calculated for a few antibiotics in groundwater. The risk assessment of antibiotics demonstrated that antibiotics with their current groundwater levels pose no human health risks, whereas only ciprofloxacin, erythromycin, flumequine, and sulfamethoxazole revealed moderate to low risks to aquatic species. The occurrence of ARGs and antibiotic resistant bacteria (ARBs) in groundwater is also not likely to pose human health risk but consumption of groundwater contaminated with ARGs and ARBs might contribute to the development of antibiotic resistance in humans. The present review also sheds light on the relationship between ARGs, antibiotics, microbial communities, and environmental factors in

groundwater, and reported a significant correlation between them. It also addresses prospects for future outlooks into further areas of relevant research.

Keywords

Antibiotic; Antibiotic resistance genes; Antibiotic resistant bacteria; Groundwater; Human health; Environmental risk

1. Introduction

Groundwater is the primary source of freshwater globally, which accounts for 97% of available freshwater on earth, while the remaining 3% is primarily surface water ([Jurado et al. 2019](#)). It is considered as the most vital source of public water supply in many regions of the world ([Szekeres et al. 2018](#)). However, anthropogenic activities, mainly industrial and agricultural, threaten groundwater quality because many pollutants can potentially reach the aquifers ([Jurado et al. 2019](#)). Some newly recognized classes of environmental contaminants such as pharmaceuticals and personal care products (PPCPs) have become a subject of scientific and public concern because of their potential bioactive properties and unknown effects on the water environment ([Pompei et al. 2019](#)). Among PPCPs, antibiotics are a widely studied class that prevents or hinders microorganism's growth, e.g., bacteria, fungi, and protozoa ([Liu et al. 2019](#)). The major antibiotics' classification and their representative drugs are displayed in **Table S1**.

Antibiotics can accumulate in water bodies because they are only partially degraded in the environment and pose ecological and human health risks ([Kümmerer 2009](#)). Various veterinary and human drugs are continuously released into the environment during manufacturing, disposal, or metabolic-excretions ([Sharma et al. 2019](#)). The antibiotics detected in groundwater from various regions vary in concentrations, suggesting different consumption patterns. The

consumption pattern of antibiotics in household and agriculture sectors differs across regions and nations in the developing world, and even antibiotics banned in developed countries are still being utilized in developing nations ([Adebawale et al. 2016](#)). The assessment of the antibiotic consumption of 76 countries found that the global consumption rate increased by 39% during the period 2000-2015 to 42.3 billion defined daily dose (DDDs) ([Klein et al. 2018](#)). In high-income countries, an increase of 6% (from 9.7 to 10.3 billion DDDs) in antibiotic consumption has been observed from 2000 to 2015, whereas a modest drop of 4% (from 26.8 to 25.7 DDDs) in the consumption rate was reported for every 1000 inhabitants/day. However, the low to middle-income countries observed an increase of 114% (from 11.4 to 24.5 billion DDDs) in antibiotic consumption, and for every 1,000 inhabitants/day, the antibiotic consumption rate showed an increase of 77% (from 7.6 to 13.5 DDDs) ([Zhang et al. 2018](#)). Therefore, per capita antibiotic consumption was reported to be high in high-income countries, but comparatively less in low- and middle-income countries, for instance, in Brazil, China, and India ([Dai et al. 2019](#)).

The continuous release of these antibiotics into groundwater select for antibiotic-resistant bacteria (ARBs) and antibiotic resistance genes (ARGs), which exhibit significant ecological and human health concerns worldwide ([Gao et al. 2020](#)). Resistant bacteria and genes when entering a healthy organism, tend to proliferate in there, increasing in number, therefore forming a resistance module against the antibiotic within the organisms ([Singer et al. 2016](#)). As a result, the curative abilities of the antibiotics are reduced. Also, other methods like sewage sludge and animal manure that are used in agricultural lands need to be looked into to control the levels of ARBs and ARGs that find their way to the healthy organisms ([Gao et al. 2012](#)).

Studies have shown that antibiotics make their way into the water reservoirs and aquatic environments due to the inability of treatment plants to remove them altogether ([Ebele et al.](#)

[2017](#)). They have been detected in various environmental matrices, especially in the aquatic environment. They possess high biological activity even at low concentrations and can cause toxic effects to human and aquatic life. It has been reported that even low-level exposure (ng/L) shows significant impacts in impairing the normal body function, especially if the dose was administered without necessity ([Ahmed et al. 2015](#), [Kumar et al. 2019](#)). Therapeutic drugs like antibiotics may retain their properties or convert into more active compounds in the aquatic environment. Their use is unlikely to be restricted owing to the benefits they provide ([Yang et al. 2017](#)). However, the absence of advanced treatment processes and the unregulated use, and sale of such products in developing countries further aggravates the problem ([Kumar et al. 2019](#)).

As a way forward, it is essential to develop a correlation between antibiotics, their metabolites, and conventional water quality parameters ([Yang et al. 2013](#)). Antibiotics' susceptibility to various chemical, biological, and physical treatment processes varies due to the diverse nature of their chemical classes. This requires further comprehensive studies to understand better their behaviour in aquatic environment and treatment processes required for abatement ([Le-Minh et al. 2010](#)). No comprehensive review on the occurrence of antibiotics, ARGs, and ARBs in groundwater has been conducted until now. In this review, we collate the knowledge on the timeline of antibiotics and resistance development, sources, consumption, occurrence, dissemination, and risk of antibiotics, ARGs, and ARBs in groundwater. The objectives were (1) to summarise the sources, exposure pathways, and factors affecting the transport and fate of antibiotics and ARGs in groundwater, (2) to investigate the contamination profiles of antibiotics and ARGs with high concentrations in groundwater through meta-analysis, (3) to understand the occurrence, distribution, and potential drivers of ARBs in groundwater, (4) to elucidate the potential relationship between antibiotics, ARGs, ARBs, and environmental

factors in groundwater, and (5) to estimate the environmental risks against different aquatic species, and human health risks linked with exposure to antibiotics, ARGs, and ARBs through groundwater consumption.

2. Methods

This systematic review is compiled in compliance with the general principles recommended in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method ([Moher et al. 2010](#)) shown in **Fig. S1**. The literature search was performed between February and December 2019, with an additional search between February and August 2020. We used PRISMA to establish the criteria for selecting 125 studies, which were extracted and analyzed using Office 2016 (Microsoft, USA), OriginPro 2016 b9.3.226 (OriginLab, USA), Prism 8.4.2 (GraphPad Software Inc., USA), Biorender (online drafting tool, <https://biorender.com/>), and SPSS Version 22 (IBM, USA). More details on the implementation steps of the PRISMA method are shown in **Text S1** of Supplementary Information.

3. History of antibiotics and resistance development

The first effective antibiotics were introduced in 1937 ([Davies and Davies 2010](#)). With the discovery of new antibiotics and their overuse throughout history, various resistance mechanisms have emerged and evolved, limiting antibiotics' application ([Ventola 2015](#)). **Fig. 1** depicts the order of discovery and resistance development for primary antibiotics. Antibiotic resistance is considered as an apocalyptic threat to living beings because of the risk of returning to the pre-antibiotics era ([Carvalho and Santos 2016](#)). The detailed reviews on antibiotics discovery and their resistance timeline have been published in many literatures ([Davies and Davies 2010](#), [Ventola 2015](#), [Danso et al. 2019](#)).

4. Sources and exposure pathways of antibiotics and ARGs in groundwater

The primary exposure pathways of antibiotics and ARGs in groundwater are displayed in **Fig. 2**. Antibiotics may enter groundwater through leaky sewerage or manure storage, wastewater effluent, landfill leachate, livestock farming, discharge from hospitals, industrial premises, and animal production facilities, and runoff from aquaculture and agricultural areas treated with manure or wastewater ([Gao et al. 2020](#), [Sui et al. 2015](#), [Chen et al. 2017](#), [He et al. 2020](#)). The occurrence of ARGs in groundwater nearby landfills provides evidence that landfills are an underappreciated hotspot of antibiotic resistance, causing ARGs dissemination via flowing contaminated groundwater ([Chen et al. 2017](#)). It was observed that long-term aquaculture activities in the Honghu Lake, China had impacted on the dissemination of some antibiotics and their associated ARGs (*tetG*, *tetQ*, and *tetX*) in the groundwater. Fishponds or aquafarms can act as reservoirs for antibiotics and ARGs, eventually reaching groundwater ([Tong et al. 2020](#)). Swine farming was reported to be a major source of ARGs spread in groundwater bodies ([Gao et al. 2020](#), [Awad et al. 2014](#), [Li et al. 2018](#), [Chee-Sanford et al. 2001](#)).

Resistant bacteria are expected to follow the route of dispersal, similar to antibiotics ([Berglund 2015](#)). This can result in resistance hotspot environments where antibiotics, ARGs, resistant bacteria, and environmental bacterial flora (potential ARBs) are mixed. Horizontal gene transfer (HGT) in such hotspots is a major factor for forming new resistant strains ([Wellington et al. 2013](#)). It is the main mechanism of ARGs transfer among the diversity of bacteria, including environmental, non-pathogenic, human pathogenic, gram-positive, and negative via mobile DNA elements such as plasmids and transposons – either with or without mobile integrons ([Aminov 2011](#)). Mobile integrons can recognize and acquire gene cassettes containing ARGs, while chromosomal integrons carry a large number of gene cassettes ([Khan et al. 2013](#)). ARG spread

due to class1 integrons in pathogens is a serious challenge to clinical therapy ([Stalder et al. 2012](#)). Humans can be exposed to resistant bacteria by numerous routes including consumption of contaminated crops, drinking contaminated groundwater, etc. In humans, resistant bacteria can spread ARGs to the human microbiome ([Wellington et al. 2013](#), [Allen et al. 2010](#)).

5. Factors affecting transport and fate of antibiotics and ARGs in groundwater

The transport and transformation of antibiotics in groundwater are primarily influenced by several environmental factors and physicochemical properties of the compounds ([Boy-Roura et al. 2018](#)). The vital factors involved in the transport and fate of antibiotics and ARGs in groundwater are portrayed in **Fig. 3**.

5.1. Reactivity factors

5.1.1. Sorption

Sorption is critical in controlling the passage of antibiotics to groundwater ([Yang et al. 2017](#)). The distribution coefficient (K_d) characterizes the sorptive exchange of antibiotics between water and solid phases. The K_d for organic compounds varies with soil type and its organic carbon-water partition coefficient (K_{OC}) ([Boy-Roura et al. 2018](#)). Antibiotics are complex molecules and their sorption behavior can be determined considering the initial concentration, ionic strength, solid-to-liquid ratio, temperature, pH, and octanol/water partition coefficient (K_{ow}) ([Loftin et al. 2008](#)).

5.1.2 Biodegradation

Antibiotics' biodegradability is determined by their molecular structure and physicochemical properties, e.g., sulfamethoxazole experienced less biodegradability than other antibiotics in microcosm studies ([Boy-Roura et al. 2018](#)). A study on anaerobic biodegradation

of antibiotics showed 85% and 62% removal of ciprofloxacin under sulfate- and nitrate-reducing environments, respectively, while sulfamethoxazole exhibited no biodegradability ([Martins et al. 2018](#)). Redox conditions also control pharmaceuticals' biodegradation in groundwater. Redox-dependent biodegradation via tank aeration experiments showed an efficient removal of doxycycline under aerobic conditions, whereas for clarithromycin, clindamycin, and roxithromycin only under anaerobic conditions ([Gartiser et al. 2007](#)). Moreover, anaerobic conditions are also reported to be prevalent in the groundwater environment, which can play a significant role in the biodegradation of antibiotics, as previously reported for WWTPs ([Zhao et al. 2020](#)).

Environmental conditions also show various influences on antibiotics' biodegradation. Sui et al. showed dissimilarity in tetracycline and chlortetracycline biodegradation at varying temperatures and pH, whereas the degradation of sulfonamides was less affected by changes in temperature and pH, signifying groundwater removal mechanisms might not be pH dependent reactions, i.e., hydrolysis ([Sui et al. 2015](#)). Microorganisms in groundwater are not abundant and diverse as compared to those present in soil ([Lapworth et al. 2012](#)). Hence, antibiotics in groundwater might undergo incomplete degradation and potentially convert to hazardous metabolites or remain unchanged in groundwater for a long time ([Sui et al. 2015](#)). The study conducted by Galán et al. highlighted the occurrence of five acetylated metabolites of sulfonamides in groundwater of Spain. It was found that sulfamerazine metabolite (N^4 -acetylsulfamerazine) occurred more frequently than their parent compound (sulfamerazine) with the maximum concentration of 18 ng/L. Furthermore, the occurrence of N^4 -acetylsulfapyridine (metabolite of sulfapyridine) was also reported to be higher in the groundwater. This study

further showed that antibiotics metabolites are more soluble and exhibit higher stability than their parent compounds ([García-Galán et al. 2011](#)).

5.1.3 Hydrochemical conditions

Hydrochemistry of groundwater plays a vital role in antibiotics' behavior in groundwater ([Menció and Mas-Pla 2019](#)). Sulfamethoxazole showed greater mobility in groundwater than ciprofloxacin under the effects of ionic strength and pH ([Chen et al. 2011](#)). Transportation of reactive components of antibiotics has significance in their existence in groundwater, thus, the retardation factor provides a quantitative idea of antibiotics' migration rate concerning groundwater flow ([Boy-Roura et al. 2018](#)).

5.2. Physicochemical factors

Selective pressure is the main contributor in HGT of ARGs and their distribution in groundwater ([Allen et al. 2010](#), [Amarasiri et al. 2019](#)). The exponential increase in production and usage of antibiotics for various purposes, including the treatment of ailments and growth enhancement, are the major causal factors of this selective pressure ([Zhang et al. 2009](#)). An increase in such selective pressure has been found to facilitate bacteria in the attainment of ARGs, further facilitating ARGs' spread ([Luo et al. 2005](#)).

The antibiotics found at low sub-inhibitory concentrations lead to an increase in ARGs' spread in the water environment ([Greenfield et al. 2018](#)). For instance, sulfamethoxazole slightly increased antibiotic resistance's transfer frequency at sub-inhibitory concentration of antibiotics in an aquatic environment ([Jutkina et al. 2018](#)).

The role of metals (e.g., Hg, Zn, Cu, and Cd) in the co-selection of antibiotic resistance might be another important factor contributing towards ARGs' spread ([Amarasiri et al. 2019](#)).

Compared with control areas, bacteria in metal-contaminated environments easily obtain antibiotic resistance phenotypes ([Zhang et al. 2009](#)). Co-selection mechanisms include co-resistance and cross-resistance. Co-resistance occurs when a single genetic element carries various resistance determinants, whereas, in cross-resistance, a single genetic determinant leads to antibiotic as well as metal resistance. Thus, metal contamination is a major contributor to antibiotic resistance and is crucial in environmental and clinical settings ([Baker-Austin et al. 2006](#)). Since the literature on fate and transport of ARGs in groundwater is limited, more comprehensive research is needed to identify the key factors influencing ARGs spread.

5.3. Hydrogeological factors

Aquifers' geological features and hydrological sources contribute to its recharge and antibiotic content ([Menció and Mas-Pla 2019](#)). Streamflow of the river is also expected to contribute to antibiotic content in wells nearby. Rainfall recharge infiltrates manure that applied as fertilizer to crops, causing pollutants to enter the vadose zone and later the aquifer. Partially saturated zones temporally retain pollutants and progressively leach them to groundwater by gravity flow after precipitation ([Boy-Roura et al. 2018](#), [K'oreje et al. 2020](#)). Rainfall is also an important factor that plays a vital role in the dissemination of ARGs in groundwater. The study conducted by Huang and colleagues reported that high-intensity rainfall drove the dissemination of tetracycline and sulfonamide ARGs to groundwater with the assistance of a stream ([Huang et al. 2019](#)).

6. Occurrence and spatial distribution of antibiotics and ARGs in groundwater: A global scenario

Studies on the occurrence and distribution of antibiotics and ARGs in the groundwater have been conducted in many regions worldwide ([Carvalho and Santos 2016](#), [Liu and Wong](#)

[2013](#)). The levels of antibiotics and ARGs in groundwater vary across the countries and amongst different regions of the same country. For developed regions, data on antibiotics and ARGs groundwater contamination are more prevalent in Europe and the United States, whereas, in developing regions, the data is more prevalent in China. In Asia, China is the only country with well-established literature on the occurrence of antibiotics and their related genes in groundwater ([Liu and Wong 2013](#)). Whereas, In the American continent, the United States has the highest number of published studies on the groundwater level of antibiotics and ARGs ([Chee-Sanford et al. 2001](#), [Koike et al. 2007](#), [Mackie et al. 2006](#)). The data on the occurrence of antibiotics in Africa is scarce due to the limited availability of analytical techniques ([K'oreje et al. 2020](#)). However, considerable data is available in Europe compared to other countries ([Carvalho and Santos 2016](#)).

This review compares the reported concentrations of sulfonamides, quinolones, macrolides, and tetracyclines antibiotics groups in groundwater globally (**Table S2**), and the occurrence of ARGs and methodologies used to detect ARGs in groundwater have also been summarized in **Table 1**. Different types of ARGs reported in the groundwater have been described in **Table S3**.

6.1. Tetracyclines and associated ARGs

Tetracyclines are a broad class of veterinary drugs extensively used worldwide ([Dai et al. 2019](#)). The application of tetracycline antibiotics to animal feed additives was first approved by the US Food and Drug Administration (FDA) in 1950 ([Daghrir and Drogui 2013](#)). It is ranked second in production and consumption worldwide, whereas ranked first in China with respect to its usage ([Kovalakova et al. 2020](#)). China and the US are the largest tetracycline producers due to

increased animal farming and their maximum use in the agricultural sector as a feed supplement to promote growth ([Dai et al. 2019](#)).

The global contamination level of tetracyclines in groundwater is shown in **Fig. 4a**. The elevated concentration of tetracyclines in groundwater was reported in the US, and Canada (America), followed by Europe and Asia (**Fig. 4a**). Among the tetracycline antibiotics, tetracycline is the most prescribed veterinary antibiotic, found with the highest concentration of approximately 500 ng/L in groundwater of the US, which were polluted by swine confinement facilities ([Mackie et al. 2006](#)). Tetracycline and its related ARGs were also observed in groundwater of Cluj-Napoca, Romania, impacted by the swine farms with no proper waste handling system ([Szekeres et al. 2018](#)). The existence of tetracycline ARGs has been found more frequently in China's groundwater because of its highest production (97,000 tons/year) in the region, followed by the US, Canada, and Spain ([Dai et al. 2019](#)).

According to Chee-Sanford et al., the presence of the tetracycline ARGs in deep wells is higher than shallow wells; this outlines that the impurities may get vertically distributed, dependent on the hydrogeology of the location. As a result, the movement and seepage of the underlying inlets cause tetracycline ARGs to enter groundwater. The consumption of groundwater contaminated with tetracyclines provides a possible way by which ARGs enter the human and animal food chains ([Chee-Sanford et al. 2001](#)). Groundwater wells next to swine production facilities contain high levels of tetracycline and associated ARGs than the wells located more distant from swine farming production facilities ([Koike et al. 2007](#)). It was found that swine production facilities are the potential source of tetracyclines and related genes in groundwater ([Szekeres et al. 2018](#), [Gao et al. 2020](#), [Tong et al. 2020](#), [Chee-Sanford et al. 2001](#), [Koike et al. 2007](#), [Mackie et al. 2006](#), [Frey et al. 2015](#), [Yao et al. 2015](#), [Tong et al. 2014](#)), other

than aquaculture ([Tong et al. 2020](#), [Yao et al. 2017](#)) and livestock production ([Dai et al. 2019](#), [Daghrir and Drogui 2013](#), [Yao et al. 2017](#), [Chen et al. 2018](#)).

6.2. Sulfonamides and associated ARGs

Sulfonamides are one of the oldest antibiotic drugs which are still widely used nowadays ([Boy-Roura et al. 2018](#)). It is one of the most extensively studied class of antibiotics and has been reported to be present at higher concentrations in several studies ([Szekeres et al. 2018](#), [Yao et al. 2015](#), [Lin et al. 2015](#), [Peng et al. 2014](#), [López-Serna et al. 2013](#)). **Fig. 4b** highlights the relative distribution levels of sulfonamides in groundwater worldwide. It was observed that the European groundwater is highly contaminated with sulfonamides. Among sulfonamides, sulfamethoxazole is currently the most detected antibiotic in groundwater, and its occurrence in the groundwater environment ranges from not detected to 1820 ng/L, displayed in **Table S2**. This extremely high concentration of sulfamethoxazole was reported in shallow groundwater of Taiwan, which was significantly impacted by anthropogenic activities, i.e., leakage of a landfill and wastewater discharge from hospitals and pharmaceutical companies ([Lin et al. 2015](#)). The occurrence of 171 unique ARGs in groundwater near municipal solid waste landfill has also been reported in which *sul1* and *sul2* (sulfonamides ARGs) were among the most detected resistance genes, after tetracycline ARGs ([Chen et al. 2017](#)). A significant abundance of sulfonamide ARGs has also been observed in groundwater nearby sewage treatment plant, and it was found that *sul1* and *sul2* had the maximum relative abundance in groundwater, compared to other water bodies in Beijing, China ([Liu et al. 2019](#)). The occurrence of sulfonamide resistance genes (*sul1* and *sul2*) has also been observed in groundwater near a swine feedlot ([Gao et al. 2020](#)). The widespread occurrence of sulfonamides in groundwater might be due to their weak sorption

tendency to the soil, low biodegradability, and fast mobility, which makes leaching of these antibiotics to groundwater easy ([Boy-Roura et al. 2018](#)).

6.3. Quinolones and associated ARGs

The quinolone class of antibiotics was one of the most recent ones to be utilized in clinical practice ([Carvalho and Santos 2016](#)). Quinolones are polar, generally amphoteric, and display low water solubility at pH 6–8 ([Boy-Roura et al. 2018](#)). They are utilized in human and veterinary drugs to treat bacterial infections ([Carvalho and Santos 2016](#)). The relative distribution levels of quinolones in global groundwater are shown in **Fig. 4c**, which depicts the level of quinolones to be high in the groundwater of Asia.

Among the quinolone class, ciprofloxacin is the most prescribed human antibiotic. Its groundwater occurrence has been reported at 14000 ng/L in samples collected from wells in villages falling within the proximity of effluent treatment plant that treats wastewater of nearly 90 bulk drug manufacturers ([Fick et al. 2009](#)). The elevated concentrations of antibiotics in groundwater of Barcelona, Spain revealed that groundwater samples were collected from the sites, recharged by natural bank filtration that receives the highest percentage of effluents from wastewater treatment plants (WWTPs) ([Lopez et al. 2015](#)). The quinolones resistance genes (*qnrB* and *qnrS*) have been detected in Brazil's unregistered housing estate wells ([Tolentino et al. 2019](#)). The study conducted in China has also reported quinolone resistance genes (*qnrA*, *qnrB*, and *qnrS*) in groundwater impacted by aqua farming ([Tong et al. 2020](#)). Although studies on groundwater contamination of quinolones are scant, the available literature suggests that quinolones have a high sorption tendency to soil, and once they reach the groundwater, their degradation rate slows down due to anaerobic conditions ([Ma et al. 2015](#)).

6.4. Macrolides and associated ARGs

Macrolides are clinically efficacious for treating common respiratory and dermal infections ([Alvarez-Elcoro and Enzler 1999](#)). Macrolides, particularly erythromycin, roxithromycin, clarithromycin, and azithromycin have been widely reported in groundwater bodies (**Table S2**). The relative distribution levels of macrolides in groundwater worldwide are summarized in **Fig. 4d**. When compared to other countries, China has the highest data for macrolides' concentration in groundwater. However, the concentrations of macrolides reported in other countries are higher than that of China, reflecting different consumption behavior and their continuous input to the groundwater or its proximity to a point-source. On the contrary, lower detection might be attributed to its less consumption or the geochemical processes, limiting its transport to the groundwater ([Carvalho and Santos 2016](#)).

The study conducted by Bartelt-Hunt et al. reported the maximum concentration of erythromycin at 2380 ng/L in shallow groundwater in the vicinity of two beef cattle amenities ([Bartelt-Hunt et al. 2011](#)). The study demonstrated that groundwater underlying some livestock wastewater impoundments is prone to pharmaceutical contamination. The data on the occurrence of azithromycin in groundwater is limited; however, the highest level of azithromycin was reported to be 1620 ng/L in confined aquifers in Spain. The main contributors of pollution in this area were losses from the sewage system and water supply network ([López-Serna et al. 2013](#)).

Macrolide ARGs have also been observed in groundwater and reported by many researchers ([Szekeres et al. 2018](#), [Chen et al. 2017](#), [Tong et al. 2020](#), [Böckelmann et al. 2009](#), [Wu et al. 2020](#)). Chen and colleagues found eight different erythromycin ARGs in groundwater nearby municipal solid waste landfills in China ([Chen et al. 2017](#)). The occurrence of macrolides in groundwater has been studied in France ([Lopez et al. 2015](#)), Germany ([Lapworth et al. 2012](#)),

and Romania ([Szekeres et al. 2018](#)). The erythromycin ARGs, e.g., *erm B*, have also been observed in groundwater from rural and urban settings in Romania, contaminated by anthropogenic activities. Macrolides' persistence in the water environment is attributable to their substantial and constant consumption in both human and veterinary medicines ([Carvalho and Santos 2016](#)).

More attention is now being focused on the presence of antibiotics in groundwater. Since the consumption of antibiotics is on the rise in animal production and human health across the world, the increased emergence of ARGs in groundwater is also an area of future research interest.

7. Occurrence of ARBs in groundwater

Groundwater serves as a primary source of drinking water for billions of people around the world ([Murphy et al. 2017](#)). Perhaps finding the sources and identifying the pathways for ARBs is critical, little is known about their proliferation in groundwater ([Gao et al. 2020](#), [Andrade et al. 2020](#)). An analysis of the occurrence of ARBs in the global groundwater samples showed that four-fifths of the isolates obtained from samples were resistant to one or more antimicrobial agents ([Andrade et al. 2020](#)). A study conducted by Akaniro and colleagues showed that about 86% of the isolates detected in groundwater samples were multidrug-resistant. The characterization and identification of these isolates showed that 30 different coliform bacteria belong to nine different genera, namely *Salmonella*, *Citrobacter*, *Klebsiella*, *Aeromonas*, *Vibrio*, *Pseudomonas*, *Enterobacter*, *Escherichia*, and *Serratia*. The *Salmonella* spp. was highly prevalent with 23.3% abundance, whereas *Enterobacter* spp. and *Pseudomonas* spp. were the isolates with the least occurrence having a prevalence of only 3.3% ([Akaniro et al. 2019](#)).

A study to investigate the prevalence of antimicrobial-resistant *Escherichia coli* (*E. coli*) isolated from the residential water wells in South Central Virginia showed that the contamination of groundwater not only poses a threat to the health of the potential users but also aggravates the problem by making treatment of the bacterial diseases difficult due presence of ARBs ([Kim et al. 2018](#)). The detection of a total of 92 strains obtained from well water showed that 65% of the bacterial strains were resistant to at least one antibiotic, and 12% showed intermediate resistance. The findings generated enough evidence to prove the prevalence of ARBs in well water, indicating that they are the reservoirs for antibiotic resistance genes ([Machado and Bordalo 2014](#)).

The analysis of groundwater samples collected from the areas with relatively higher anthropogenic pressure showed a distinct bacterial community. The taxonomic classification revealed the presence of 26 different phyla, of which the most abundant ones include *Proteobacteria*, *Bacteroidetes*, *Firmicutes*, and *Actinobacteria* ([Szekeres et al. 2018](#)). The physicochemical analysis of the groundwater samples collected from Uttar Pradesh, India showed it not to be suitable for drinking purposes. The microbial analysis showed a high antibiotic resistance in *E. coli* against amoxicillin and erythromycin. Moreover, 12 isolates (75%) were found to be multidrug-resistant with index >20% ([Sharma et al. 2017](#)). Researchers tested the occurrence of *Enterococcus* species resistant to various antibiotics in groundwater samples (n = 22). Results indicated that 8.5 to 100% of the isolates were resistant to ampicillin, penicillin G, and vancomycin, while 4.5 to 9.8% of the isolates were resistant to kanamycin and chloramphenicol ([Ateba et al. 2013](#)). The isolates of vancomycin ARGs indicated that these could pose a severe challenge to humans since infections caused by them are tough to cure ([Ateba and Maribeng 2011](#)).

A total of 101 isolates from groundwater were classified and tested for their resistance towards four commonly used antibiotics (chloramphenicol, tetracycline, neomycin, penicillin). The study found 77 isolates to be resistant to penicillin, 32 isolate to tetracycline, and 18 to chloramphenicol ([El-Zanfaly et al. 1988](#)). A study conducted in the US isolated *Enterococcus spp.* which was resistant to erythromycin, tetracycline, and vancomycin in groundwater samples downstream and upstream of swine feedlots. It was found that resistance to tetracycline and clindamycin was higher at the downgradient groundwater than that at the up-gradient due to a malfunctioning septic tank up-gradient ([Sapkota et al. 2007](#)).

Dissemination and transport of bacteria into and within groundwater can be affected by various environmental factors, such as characteristics of soil, pH, temperature, nature, and magnitude of organic matter, redox potential, water table depth, salinity, and other physiological parameters ([Sanganyado and Gwenzi 2019](#)). Chen and co-workers observed inhibition of tetracycline bioavailability to *E. coli* while exploring whole-cell bioreporter and observed complexation of tetracycline with dissolved organic matter, which led to a decrease in antibiotic resistance genes' expression ([Chen et al. 2015](#)). A study conducted to investigate the cell morphology found tetracycline susceptible bacteria to be less mobile in comparison to manure derived tetracycline-resistant *E. coli* ([Walczak et al. 2011](#)). Another research conducted on cell morphology suggests that bacteria's mobility has increased due to the enhanced forces of repulsion between the tetracycline-resistant *E. coli* and the sand quartz's anions. Despite both the strains being similar in sizes, tetracycline-resistant *E. coli* proved to be more hydrophilic, suggesting its ability to transport planktons in groundwater ([Sanganyado and Gwenzi 2019](#), [Walczak et al. 2011](#)).

Nevertheless, there is a requirement for systematic studies on the impact of human activities, such as septic tank, livestock production facilities, and environmental factors on transport and dissemination of ARBs in groundwater, especially in developing nations where consumption of untreated drinking water is common.

8. Nexus among antibiotics, microbial communities, environmental factors, and ARGs in groundwater

8.1. Antibiotics vs. ARGs

The correlation among antibiotics, ARGs, microbial community, and environmental factors in groundwater reported in various studies have been summarized in **Table 2**. Many studies have confirmed that antibiotics play an important role in inducing ARGs ([Qiao et al. 2018](#), [Rodriguez-Mozaz et al. 2015](#)). Nevertheless, the existence of an expected correlation among antibiotics and ARGs have not reached a uniform conclusion. The study conducted by Liu et al. found significant positive correlations between the concentrations of antibiotics and their corresponding ARGs in groundwater ([Liu et al. 2019](#)). It was further revealed in other studies that antibiotics dispersion could contribute to disseminating resistance genes ([Rodriguez-Mozaz et al. 2015](#), [Wang et al. 2018](#)). The tetracycline ARGs (*tetG*, *tetM*, and *tetO*) in village groundwater have been reported to be significantly correlated with the level of tetracycline at a given site. It was suggested that HGT might play a role in ARGs dissemination from wastewater to groundwater bodies ([Li et al. 2018](#)). Contrary to the above-cited literature, few studies also observed a zero correlation between antibiotic concentrations and ARG abundance in groundwater ([Szekeress et al. 2018](#), [Tong et al. 2020](#)).

8.2. ARGs vs. environmental factors

The list of ARGs correlated with environmental factors is summarized in **Table 2**. A strong correlation exists between the abundance of ARGs and environmental factors. The study conducted by Gao et al. observed 74.4% of the variance for environmental factors, and it was found that nutritional factors, i.e., total organic carbon (TOC), total phosphate (TP), total nitrogen (TN), and ammonia nitrogen (NH₄-N) influenced ARGs more than physicochemical parameters, i.e., temperature, pH, and dissolved oxygen (DO) ([Gao et al. 2020](#)). A study reported a significant correlation between ARGs and water quality parameters such as DO, NH₄-N, and TOC, which may be significantly affected by anthropogenic activities ([Wu et al. 2020](#)). Another study indicated that the detected ARGs and their correlation with environmental factors were likely driven via surface-groundwater interaction. Hence, the relationship between ARGs and environmental factors reflects the potential impacts of environmental factors on the dissemination of ARGs in groundwater ([Tong et al. 2020](#)).

8.3. ARGs vs. microbial community

The contamination profiles of ARGs and microbial communities in groundwater have been investigated in a few studies ([Szekeres et al. 2018](#), [Gao et al. 2020](#), [Tong et al. 2020](#)). A high correlation between ARGs and 16S rRNA gene was observed, reflecting that pig farming causes ARGs contamination in groundwater primarily through elevating the absolute abundance of microbes harboring ARGs ([Gao et al. 2020](#)). A strong connection was observed between *ermA* and *Faecalibacterium* genera in the groundwater of Cluj-Napoca, Romania. The study reported that resistance genes harbored the abundance of *Faecalibacterium* in groundwater to 2.8% ([Szekeres et al. 2018](#)). The occurrence of *ermB* and *tetO* in the artificial groundwater recharge system also exhibited a correlation with the abundance of *Enterococci* (human pathogen with

long survival capacity in water). It was observed that 34.6% of *Enterococcus spp.* isolates were tetracycline-resistant and 24.8% to erythromycin ([Böckelmann et al. 2009](#)). A significant correlation was also observed between the ARGs profile and microbial community. It was revealed that microbial taxa might be a potential host of ARGs ([Chen et al. 2017](#)).

9. Ecological and human health implications of antibiotics, ARGs and ARBs

9.1. Environmental risks of antibiotics

Assessment of environmental risks has become imperative to estimate pharmaceuticals' potential impacts on particular species in different environmental matrices ([Ashfaq et al. 2017](#)). In this review, ecological risk assessment of antibiotics is performed following the EU recognized standard method as a risk quotient approach by using measured environmental concentrations (MEC) and predicted no-effect concentrations (PNEC) (Eq. (1)). The value of MEC was obtained from the available literature (**Table S2**). In this review, the highest reported concentrations of antibiotics were used to evaluate the worst-case scenario of ecological risk associated with groundwater contamination. According to generally used ranking criteria, the value of $HQ < 0.1$ indicates a negligible risk to aquatic organisms, $0.1 \leq HQ < 1$ implies moderate risk, and $HQ \geq 1$ depicts high risk ([Zhu et al. 2019](#)).

$$HQ = \frac{MEC}{PNEC} \quad (1)$$

Limited studies have reported the environmental risks associated with antibiotics groundwater contamination ([Jurado et al. 2019](#), [Sharma et al. 2019](#), [Mutiyaar and Mittal 2014](#)). Here, the acute toxicity data of the ten antibiotics (tetracycline, oxytetracycline, sulfamethoxazole, ciprofloxacin, norfloxacin, ofloxacin, enrofloxacin, flumequine, erythromycin, and clarithromycin) using different organisms *P. subcapitata* (algae), *D. magna* (invertebrate),

and *V. fischeri* (bacteria) was collected from the literature and toxicity database, i.e., ECOTOX ([Zhou et al. 2019](#)). The summary of acute EC50 values is given in **Table S4**. An assessment factor with the value of 1000 was used for acute toxicity ([Vestel et al. 2016](#)). PNECs were calculated by taking the acute EC50 value of antibiotics for the targeted species and dividing it by an assessment factor of 1000. The boxplots in **Fig 5** show that HQ for most of antibiotics is very low, indicating no risk to aquatic species (**Table S5**). However, ciprofloxacin and sulfamethoxazole pose medium to low risk (HQ= 0.2) to *P. subcapitata* (**Fig. 5a**) while erythromycin exhibits moderate risk (HQ= 0.5) to *D. magna* (**Fig. 5b**). Furthermore, flumequine also shows moderate risk (HQ= 0.59) to *V. fischeri* (**Fig. 5c**).

The HQ estimated for different pharmaceuticals in Spain's groundwater also indicated the negligible risk to aquatic organisms and for erythromycin, it was reported as 0.46, which had medium to low risk for *B. calyciflorus* and *C. dubia* ([Jurado et al. 2019](#)). Another study also calculated HQ for groundwater and reported minor to high risk for ciprofloxacin (HQ= 0.1 to 23) ([Jurado et al. 2019](#)). The contaminated groundwater is not likely to be the natural habitat of these species (bacteria, algae, and invertebrates); thus, there is no direct exposure to antibiotic residues. Although the occurrence of antibiotics in groundwater did not pose a high risk compared to other environmental matrices, their persistence in groundwater could promote resistance in bacteria. Scientific studies should be conducted to evaluate the chronic exposure of various antibiotics to humans via groundwater consumption. Moreover, the remediation of contaminated groundwater is complicated because contaminants may persist in the underlying aquifers for an extended period ([Jurado et al. 2019](#)). Hence, monitoring and control of antibiotics pollution in groundwater should be the prime concern.

Among all the antibiotics consumed globally, only three antibiotics from the macrolide class (azithromycin, clarithromycin, and erythromycin) were included in the first European Union watch list of priority contaminants (Decision 2015/495) ([Szekeres et al. 2018](#)). Whereas, in the second Watch List, only two antibiotics, amoxicillin and ciprofloxacin were categorized as priority pollutants for monitoring, since these antibiotics are likely to reach groundwater bodies ([Boy-Roura et al. 2018](#)).

Moreover, owing to the diverse nature of pharmaceuticals and their anticipated risk to an aquatic environment, the Environmental Agency for England and Wales also proposed a ranking system for ten pollutants, including only two antibiotics (trimethoprim and sulfamethoxazole). This categorization was based on their persistence, bioaccumulation, toxicity, and occurrence from existing literature in various regions of the world and the availability of appropriate analytical methods for their estimation ([Ebele et al. 2017](#)).

9.2. Human health risk assessment of antibiotics

This review estimated the potential health risk of nine antibiotics (ciprofloxacin, norfloxacin, sulfamethoxazole, sulfamethazine, clarithromycin, roxithromycin, azithromycin, tetracycline, and oxytetracycline) which have been frequently reported in groundwater of various regions in the world. The risk quotient (RQ) for each antibiotic was calculated by dividing the maximum measured concentration of antibiotics in groundwater by drinking water equivalent level (DWEL) of seven categories of age (from 1 year to > 21 years) (Eq. (2)).

$$RQ = \frac{MC}{DWEL} \quad (2)$$

The guidelines from the World Health Organization (WHO, 2011) were followed to calculate DWEL (**Table S5**) (Eq. (3)).

$$DWEL = \frac{ADI \times BW}{DWI \times AB \times FOE} \quad (3)$$

where ADI is the acceptable daily intake (the dose which can be consumed daily over a lifetime having an insignificant risk of adverse effects) (**Table S6**), BW is the body weight (kg), DWI is the volume of water drank per diem (**Table S7**), AB represents the portion of the substance ingested through consumption of water (1 for pharmaceutical drugs), and FOE is the exposure frequency (0.96) ([Sharma et al. 2019](#)). The RQ value >1 indicates the possibility of human health risk, whereas a more comprehensive evaluation is required if the value ranges between 0.2 – 1. The RQ value ≤ 0.2 is of no appreciable risk to human health ([Sharma et al. 2019](#), [Yang et al. 2017](#)).

The RQ calculated in this review for various antibiotics has been presented in **Fig. 6**. Ciprofloxacin and azithromycin depicted relatively high human health risk as compared to other antibiotics. Among different age groups, children (1-11 years) had a higher risk, followed by adults (>20 years) and adolescents (11-21 years). Nevertheless, it is clear from the figure that potential health risk for all nine investigated antibiotics in groundwater is well below the 0.2 RQ guideline. This demonstrates that these antibiotics are unlikely to pose human health risks via drinking water consumption (**Table S8**). Many other studies have also estimated the level of human health risk associated with groundwater and reported negligible risk to human health ([Sharma et al. 2019](#), [Riva et al. 2018](#)). Therefore, the human health risk assessment reported in studies to date showed that the concentration of different antibiotics present in groundwater is unlikely to pose potential risks to human health.

9.3. Human health implications of ARGs and ARBs

The occurrence of antibiotics, ARGs, and ARBs in drinking water might disrupt the gastrointestinal microbiota balance and further affect human health ([Becattini et al. 2016](#)).

Nevertheless, it is still unclear whether ARBs and ARGs in groundwater have any substantial impact on humans' exposure to ARBs. The human health risks are expected to be more distinct in developing nations than developed ones. Developing countries have inadequate health care systems, characterized by an absence of necessary facilities and medications, and elevated ailment burden, which sequentially demands frequent use of antibiotics ([Gwenzi et al. 2018](#)). The estimation of the human exposure risks to veterinary antibiotics, ARGs, and bacteria like *Acinetobacter* in the groundwater of Jianzhong, China, found no immediate human health risks. The study observed that an estimate of 10 billion magnitude CFU of *Acinetobacter* would be ingested through drinking groundwater by a resident per day. However, it is still unclear whether environmental ARBs and ARGs in groundwater have any significant impact on human exposure to pathogenic ARBs ([Gao et al. 2020](#)).

The study investigated the potential human health risks of antibiotic resistance in drinking water reported that the consumption of drinking water polluted with ARBs and ARGs might contribute to the growth of antibiotic resistance in humans ([Sanganyado and Gwenzi 2019](#)). Antibiotic resistance might result in enhanced virulence, pathogenicity, disease outbreaks, and transmission, which leads to prolonged morbidity and mortality ([Berendonk et al. 2015](#)). Since evidence on the implications of antibiotic resistance in the groundwater to human health remains limited, several knowledge gaps still exist. It calls for improvement in the risk assessment methodologies to address the potential link between antibiotic resistance and other environmental stressors.

10. Research challenges and future perspectives

In recent years, considerable concern has been made to promote the prudent use of antibiotics and control antibiotics resistance in aquatic environment. But due to the lack of data

and understanding on fate and potential risks of antibiotics occurrence in the groundwater, their long term impacts in ecological environment are not clearly understood. Therefore, the inescapable consequences of these antibiotics, ARGs, and ARBs have become a global challenge. Hence, systematic studies should be conducted on the impacts of human activities and environmental factors associated with the transport of antibiotics and ARGs in groundwater. In this regard, special consideration should be taken for developing regions where consumption of untreated drinking water is common, and data is scant due to limited research facilities.

The occurrence of antibiotics metabolites has been reported to be higher than parent compounds in groundwater because of their higher solubility and persistence ([García-Galán et al. 2011](#)). Considering the inevitable use of antibiotics, and limited availability of data on hazardous effects of antibiotics metabolites, this review highlights the importance of conducting studies to monitor and find methodologies to diminish the groundwater adulteration of antibiotics and their metabolites. The occurrence of ARGs in groundwater has not been screened in many developing and developed countries, except China and the US. Hence, a comprehensive, regional, and countrywide groundwater survey should be conducted to assess ARGs' occurrence and distribution in groundwater.

Moreover, mathematical modeling has played an essential role in improving our understanding of antibiotic resistance. Various studies have conducted modeling research to evaluate factors influencing the dissemination of resistant bacteria in the environment ([Greenfield et al. 2018](#), [Smith et al. 2002](#), [Niewiadomska et al. 2019](#)). However, with rising concern about the spread of antibiotics resistance and increased use of mathematical models to better understand the mechanisms behind ARGs spread, there is a need for the development of an advanced framework that can help guide model choice for future analyses. Furthermore, the fate

and transport of ARGs should be quantitatively assessed through mathematical modeling approaches that consider ARGs replication and HGT propagation dynamics, as well as natural attenuation mechanisms (e.g., hydrolysis, photolysis, and ARG sedimentation). The evidence about the dose-response curves and exposure assessment data associated with ARBs and ARGs in groundwater is not developed yet. Therefore, Quantitative microbial risk assessment (QMRA) has been recommended as a suitable method to evaluate and quantify the human health risk ([Amarasiri et al. 2019](#)).

Certain studies offered findings on the presence of antibiotics in groundwater, yet few emphasized ARGs' existence and the processes that influence their existence in the groundwater. These investigations ignore the detailed analysis of the hydrogeological context of the research and the presence of antibiotics and ARGs within groundwater using a regional hydrogeological point of view (which is essential to deal with management strategies). Because of this complexity, the status and entry of antibiotics into groundwater supplies are not clearly understood. Comprehensive research is required to understand better hydrogeology's effect on the distribution of antibiotics and the spread of ARGs in groundwater.

In the extensive research that has been undertaken by various institutions, it is noteworthy to mention that environmental reservoirs that house most of the groundwater are on the rise to be viewed as one of the most critical hotspots for various pathogens gaining antibiotic resistance. Therefore, prescription of antibiotics can be rationalized based on the detection and identification of bacterial pathogens through the point of care (POC), thus can help in regulating and reducing inappropriate use of antibiotics, which can further enhance and prolong the effectiveness of such antibiotics. As the field of POC detection tests is growing and advancing, reliance on empirical antimicrobial treatment is losing its efficacy; therefore, quick POC tests should be employed for

bacterial infections' diagnosis before antimicrobials are prescribed to patients, improving the results and limiting the development of antimicrobial resistance ([Reali et al. 2019](#)).

The occurrence of ARBs and ARGs are reported in urban watersheds without point source adulteration, signifying that naturally produced antimicrobials might act as selective agents for the resistance of antibiotics in the watersheds ([Low et al. 2016](#)). Therefore, any water that has been linked with agricultural production units, WWTPs, aquafarming, swine production facilities, or hospital waste, where antibiotics are highly used, should be extensively examined before reusing it. Moreover, in developing regions, due to the lack of proper waste handling systems, comprehensive point and nonpoint source based monitoring of the groundwater and watersheds should be introduced to reap the benefits of good health.

11. Conclusions and take home message

The current review provides comprehensive information about the occurrence pattern of most frequently detected antibiotics, ARGs, and ARBs in groundwater globally. The reviewed studies suggested sulfonamides, macrolides, quinolones, and tetracycline classes of antibiotics to be more prevalent in groundwater bodies, owing to their significance in human and veterinary medicines and persistence in groundwater. Although these antibiotics classes are widely detected in groundwater, the data on the occurrence of sulfamethoxazole, ciprofloxacin, and tetracyclines is more prevalent in various regions of the world. The literature on the occurrence of tetracyclines and sulfonamides resistance genes is more abundant, reflecting their continuous input in the groundwater. It was found that groundwater lying in nearby proximity of point sources such as landfill leachates, wastewater discharge, aquaculture runoff, discharge from hospitals or drug manufacturing industries, and discharge from swine production facilities are more contaminated with antibiotics residues, resistance genes, and resistant bacteria. However, it

was observed that intensive swine farming is a significant source contributing to ARGs and ARBs in groundwater.

This review further collates the knowledge on the correlation between antibiotics, ARGs, environmental factors, and the microbial community. The relationship between ARGs and environmental factors reflects the potential impacts of environmental factors on disseminating ARGs in groundwater. The nexus between ARGs profile and microbial community revealed that microbial taxa might be the potential host of ARGs. The relationship between antibiotics and ARGs suggests that antibiotics dispersion in groundwater leads to the enhanced dissemination of ARGs in groundwater.

To evaluate the worst-case scenario, the highest reported antibiotics values were selected from published literature to estimate ecological and human health risks. The risk assessment showed the negligible risk to human health and medium to moderate ecological risks to antibiotics were estimated to aquatic species. Therefore, human health and environmental risk from exposure to antibiotics may be of concern, especially because groundwater is an essential source of drinking water in many regions of the world.

Considering the inescapable use of antibiotics, this review highlights the point and non-point source based monitoring to reduce the groundwater adulteration with antibiotics, ARGs and ARBs and support future regulatory measures.

Conflicts of interest

The authors declare no conflict of interest.

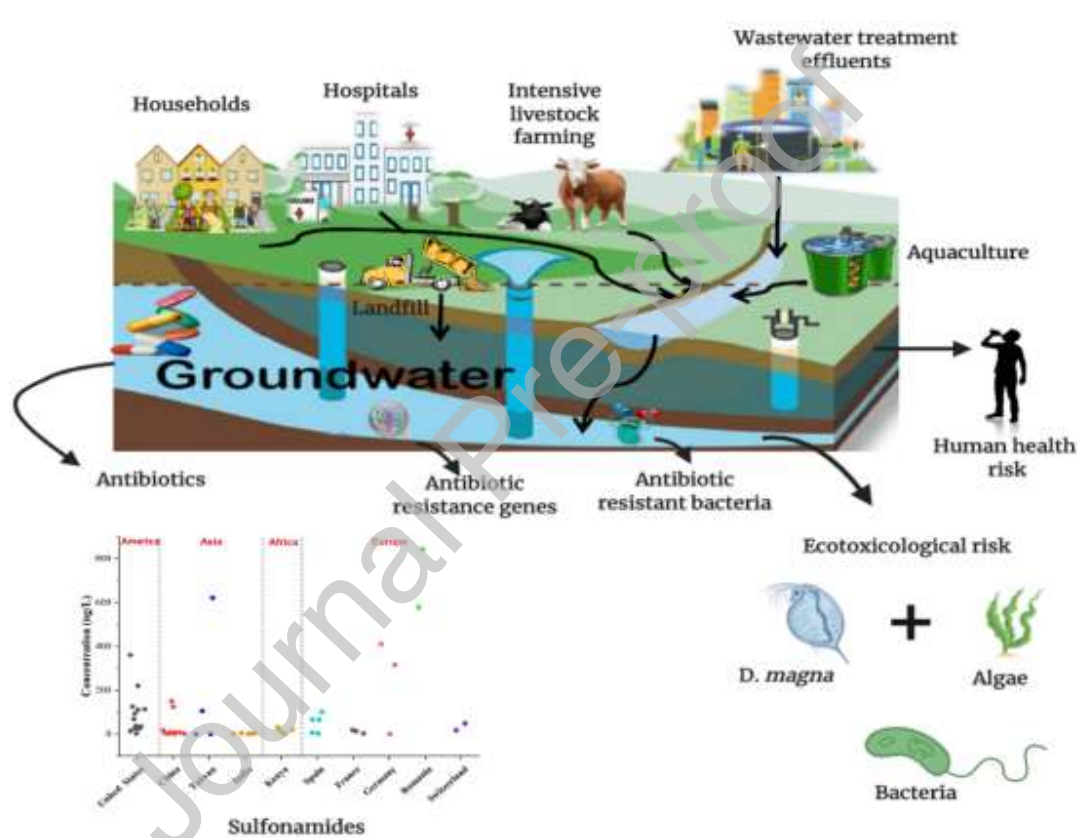
Acknowledgments

The present study was supported by China Post-Doctoral Research Funds (2019M650356) to Muhammad Junaid.

Conflict of interest

The authors declare no conflict of interest.

Graphical abstract



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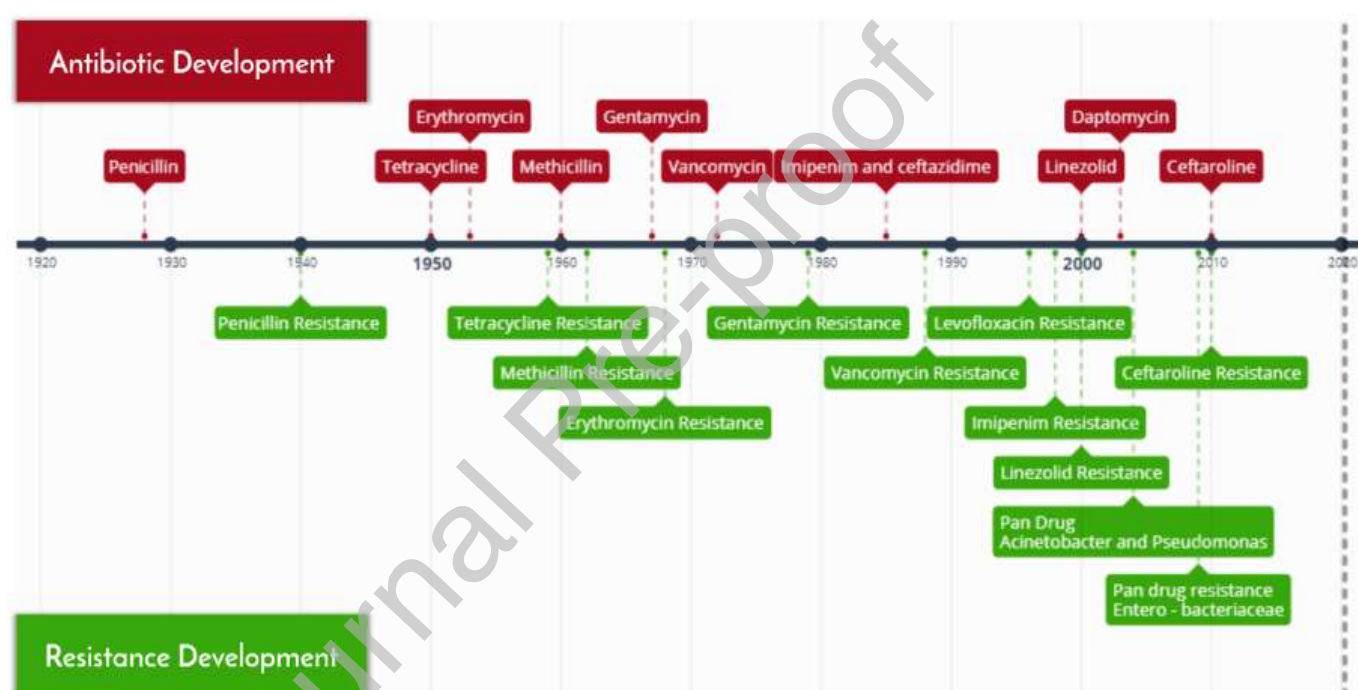


Fig. 1. History of antibiotics discovery and antibiotics resistance development. The timeline of antibiotic discovery (red boxes) and antibiotics resistance development (green boxes) are presented chronologically. Before penicillin discovery, the pre-antibiotic era was considered to be the dark-age. The halcyon years started after 1950 when most of the antibiotics used today were discovered. However, no new antibiotic has been discovered after 2010 which is the main reason for resistance development in bacteria ([Singh et al. 2020](#), [Ventola 2015](#)).

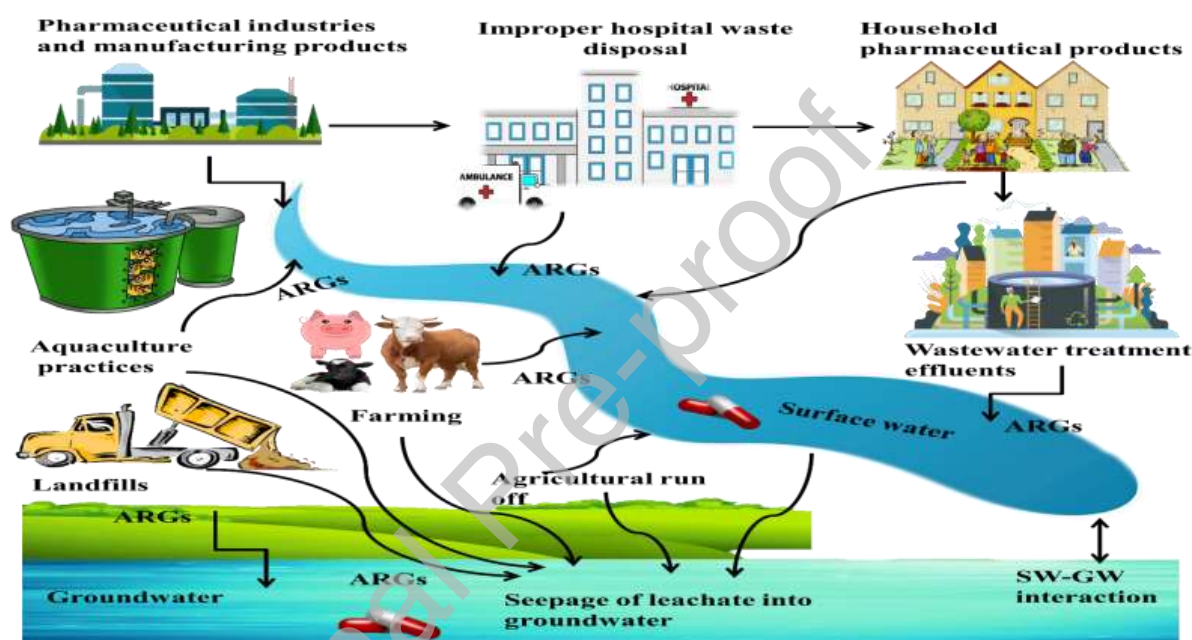


Fig. 2. Sources and exposure pathways of antibiotics and antibiotic resistance genes (ARGs) dissemination in groundwater. Human and veterinary antibiotics are released into the environment by their extensive use and disposal practices. Residues of antibiotics enter into sewage systems through effluents from medical facilities, pharmacies, excretion via animals and humans. Both treated and untreated wastewater, agricultural runoff, landfills, runoff from aquaculture facilities becomes an entry route for antibiotics into the aquatic environment and leach into groundwater. Their persistence in groundwater for a long time develop resistance in bacteria. The aquaculture activities, landfill leachates, swine farming, hospital effluents, and wastewater discharge are the major sources of ARGs dissemination in groundwater.

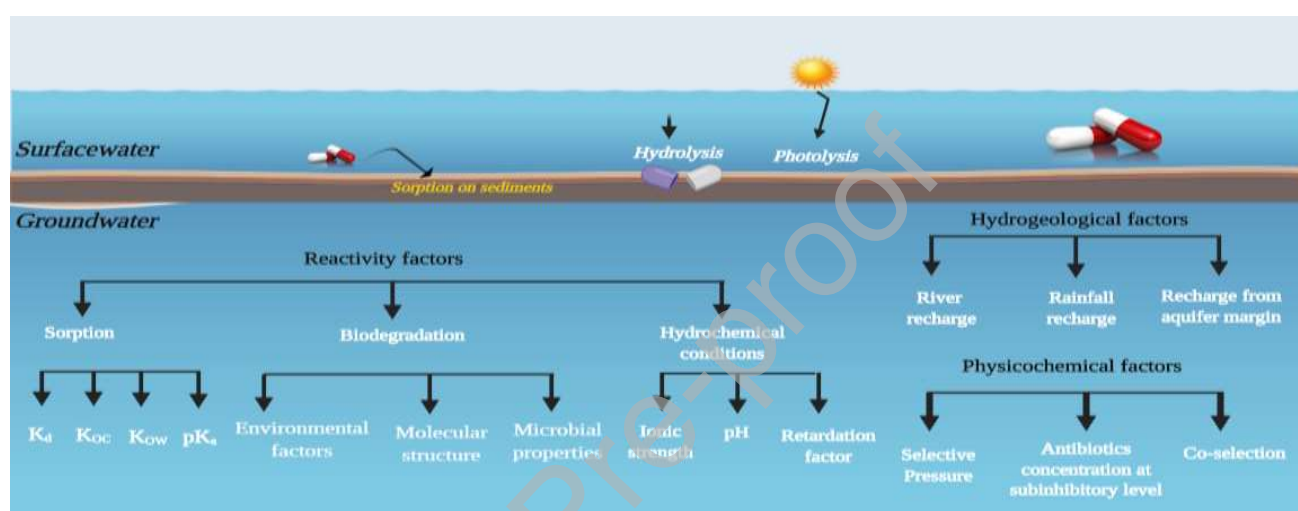


Fig. 3. Factors affecting the fate and transport of antibiotics and ARGs in groundwater. Antibiotics transport in groundwater mainly depends on their sorption on rock/sediments. The sorption behavior of antibiotics is determined by various physicochemical properties of the antibiotics, the distribution coefficient (K_d), carbon-water partition coefficient (K_{OC}), dissociation constant (pK_a), octanol/water partition coefficient (K_{ow}), and the environmental conditions (e.g. pH, temperature, infiltration rate). Besides sorption, biodegradability also affects the fate of antibiotics in groundwater. For example, as contaminated water percolates through the soil profile and reaches the aquifer, the biodegradability slows down due to the limited existence of microbial populations, nutrients, and oxygen. The incomplete degradation of antibiotics potentially changes into hazardous metabolites, which might persist in groundwater for a longer period of time. **Environmental factors, microbial properties and molecular structure of antibiotic influence the degradation of antibiotics in groundwater.** Lastly, climatic conditions such as rainfall frequency and intensity, groundwater recharge, and dilution influence the mobility of antibiotics in groundwater. High-intensity rainfall also promotes the dissemination of ARGs in groundwater. The physicochemical factors i.e. selective pressure, the occurrence of antibiotics at a subinhibitory level, and co-selection of antibiotics with metal might be the key factors that influence the ARGs in groundwater.

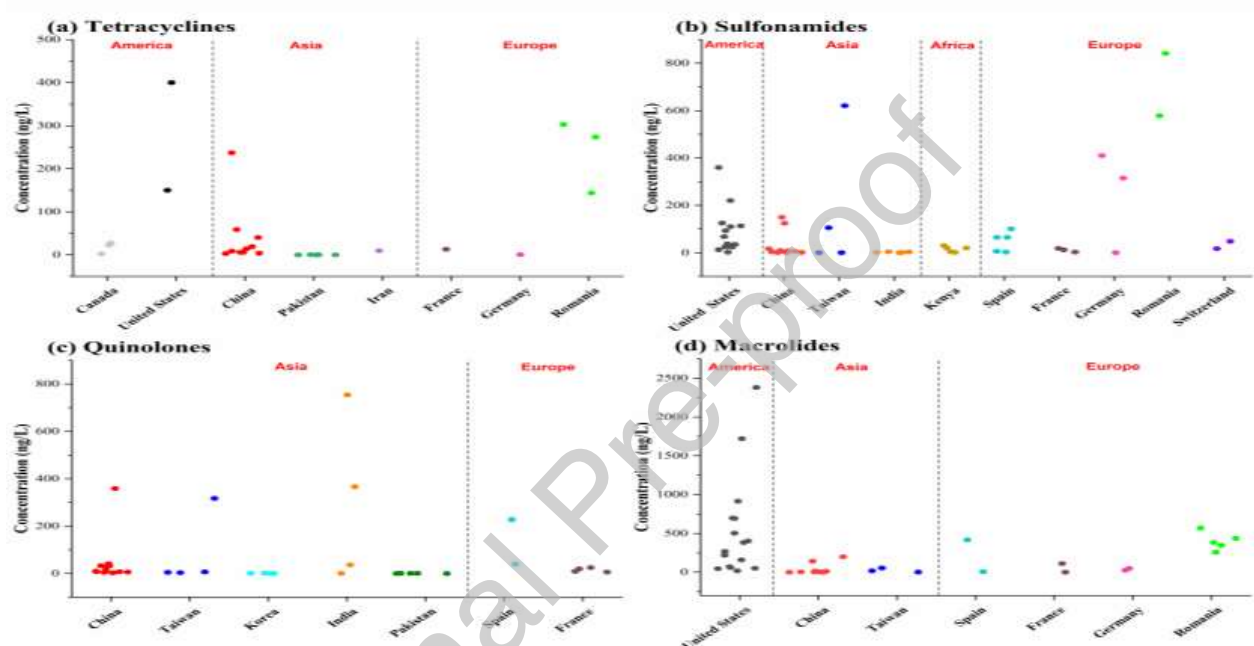


Fig. 4. Relative distribution levels of main antibiotic classes in groundwater worldwide. The groundwater levels (ng/L) of (a) tetracyclines, (b) sulfonamides, (c) quinolones, and (d) macrolides reported in different countries. Tetracyclines include tetracycline, chlortetracycline, and oxytetracycline. Sulfonamides include Sulfadiazine, Sulfamethazine, Sulfamethoxazole, and Sulfadimethoxine. Quinolones include ciprofloxacin, ofloxacin, norfloxacin, enrofloxacin, and flumequine. Macrolides include erythromycin, clarithromycin, roxithromycin, and azithromycin. The observed range of antibiotics occurrence in groundwater varies from country to country. The varying occurrence of antibiotics in groundwater around the globe is due to changing consumption patterns, groundwater sample size, well-depth, and its proximity to point source.

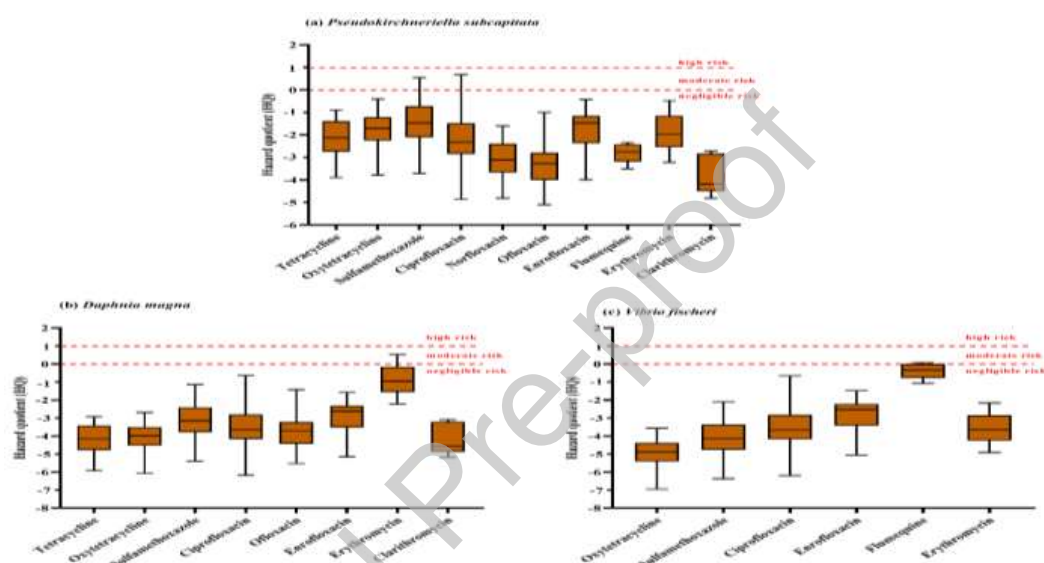


Fig. 5. Environmental risk of antibiotics in groundwater. The boxplots show the hazard quotient (log10 values) of targeted antibiotics in groundwater concerning their acute toxicity for (a) algae (*P. subcapitata*), (b) invertebrates (*D. magna*), and (c) bacteria (*V. fischeri*). Due to limited ecotoxicity data, HQ of antibiotics can only be calculated against a few species. The environmental risk of most of the antibiotics was less than 0.01, which indicates no considerable ecological risk. However, ciprofloxacin and sulfamethoxazole pose medium to low risk (HQ= 0.2) to *P. subcapitata* (Fig.5a), while erythromycin exhibits moderate risk (HQ= 0.5) to *D. magna* (Fig. 5b). Furthermore, flumequine also shows moderate risk (HQ= 0.59) to *V. fischeri* (Fig. 5c). The measured environmental concentrations (MEC) used to calculate ecological risks were taken from published literature, given in Table S2. According to generally used ranking criteria, the value of HQ<0.1 indicates a negligible risk to aquatic organisms, 0.1≤HQ 1 implies moderate risk, and HQ≥1 depicts high risk (Zhu et al. 2019).

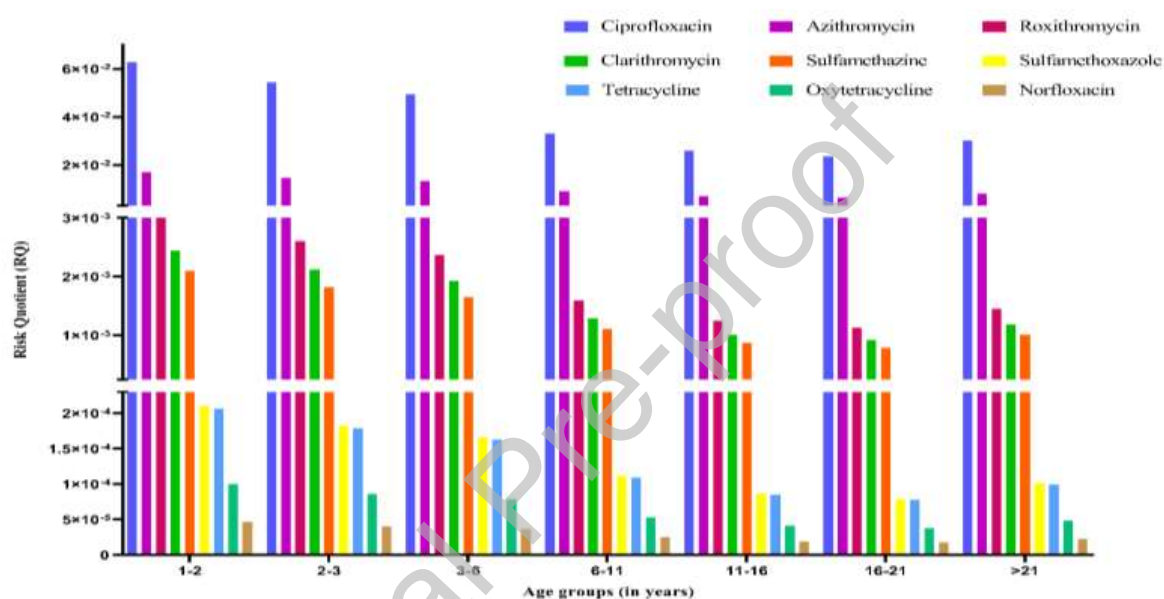


Fig. 6. Human health risks of antibiotics in groundwater. The grouped bar graphs show the human health risk quotient (RQ) of various age groups for extensively used antibiotics in groundwater. The risk quotient of targeted antibiotics in groundwater was less than 0.2 RQ guidelines indicating negligible risks for humans. Ciprofloxacin and azithromycin showed comparatively high risk than other antibiotics. Among different age groups, children posed a high risk of antibiotics through groundwater intake as compared to adults and adolescents. The acceptable daily intake (ADI) values used to calculate human health risks are given in Table S6. The RQ value >1 indicates the possibility of human health risk, whereas a more comprehensive evaluation is required if the value ranges between 0.2 – 1. The RQ value ≤ 0.2 is of no appreciable risk to human health (Sharma et al. 2019).

Table 1. Worldwide occurrence of antibiotic resistance genes (ARGs) in groundwater.

Country	Region	Quantification method	No. of ARGs detected	Antibiotic-resistant genes (ARGs)	References
Belgium	Torreele	Real-time quantitative PCR (qPCR)	2	<i>tetO</i> , <i>ermB</i>	(Böckelmann et al. 2009)
Canada	Ontario	Real-time quantitative PCR (qPCR)	1	<i>TetO</i>	(Frey et al. 2015)
Brazil	São José do Rio Preto	16S rDNA	3	<i>tetA</i> , <i>qnrS</i> and <i>qnrB</i> genes	(Tolentino et al. 2019)
Canada	Ontario	Real-time quantitative PCR (qPCR)	1	<i>TetO</i>	(Frey et al. 2015)
China	Xiamen	High throughput quantitative PCR (HT-qPCR)	171	<i>cmlA1</i> , <i>cmx</i> , <i>ampC</i> , <i>bla1</i> , <i>mecA</i> , <i>ermA</i> , <i>ermB</i> , <i>ermC</i> , <i>ermF</i> , <i>ermK</i> , <i>ermT</i> , <i>ermX</i> , <i>ermY</i> , <i>mefA</i> , <i>dfrA1</i> , <i>dfrA2</i> , <i>folA</i> , <i>sul1</i> , <i>sul2</i> , <i>sulA/folP-01</i> , <i>sulA/folP-03</i> , <i>tetA</i> , <i>tetB</i> , <i>tetC</i> , <i>tetD</i> , <i>tetE</i> , <i>tetG</i> , <i>tetH</i> , <i>tetK</i> , <i>tetL</i> , <i>tetM</i> , <i>tetP</i> , <i>tetQ</i> , <i>tetR</i> , <i>tetS</i> , <i>tetT</i> , and <i>tetX</i> etc.	(Chen et al. 2017)
China	Shunyi	Real-time quantitative	6	<i>tetG</i> , <i>tetM</i> , <i>tetO</i> , <i>qnrA</i> , <i>sul1</i> , and <i>sul2</i>	(Li et al. 2018)
China	Changping		6	<i>tetG</i> , <i>tetM</i> , <i>tetO</i> , <i>qnrA</i> , <i>sul1</i> , and <i>sul2</i>	
China	Daxing		6	<i>tetG</i> , <i>tetM</i> , <i>tetO</i> , <i>qnrA</i> , <i>sul1</i> , and	

		PCR (qPCR)		<i>sul2</i>	
China	Ninghe		6	<i>tetG, tetM, tetO, qnrA, sul1, and sul2</i>	
China	Anping		6	<i>tetG, tetM, tetO, qnrA, sul1, and sul2</i>	
China	Guangyang		6	<i>tetG, tetM, tetO, qnrA, sul1, and sul2</i>	
China	Xiqing		6	<i>tetG, tetM, tetO, qnrA, sul1, and sul2</i>	
China	Beijing	Real-time quantitative PCR (qPCR)	9	<i>sul1, sul2, tetA, tetB, tetC, tetM, tetW, qnrS, and ermB</i>	(Liu et al. 2019)
China	Maozhou	High throughput quantitative PCR (HT-qPCR)	127	<i>sul1, sul2, aadA1, ermA, cmlA, tetC, and tetO etc.</i>	(Wu et al. 2020)
China	Jianzhong, Wentang	Real-time quantitative PCR (qPCR)	19	<i>IntI1, IntI2, sul1, sul2, tetC, tetG, tetH, tetO, tetW, tetBP, qnrD, qnrS, floR, cmlA, cfr, fexA, ereA, aadA, blaTEM</i>	(Gao et al. 2020)
China	Honghu Lake	Real-time quantitative PCR (qPCR)	27	<i>ermB, ermC, ermF, ermJ, ermK, ermX, qnrA, qnrB, qnrS, sul1, sul2, sul3, tetA, tetB, tetC, tetD, tetE, tetG, tetO, tetQ, tetT, tetW, tetH, tetL, tetM, tetX, and intI1</i>	(Tong et al. 2020)
China	Zhongshan	Real-time quantitative PCR (qPCR)	3	<i>tetA, tetB, tetC, tetD, tetE, tetG, tetH, tetK, tetL, tetY, tetZ, tetA-P,</i>	(Huang et al. 2019)

				<i>tetM, tetO, tetQ, tetS, tetW, tetX and sul1</i>	
Germany		Real-time quantitative PCR (qPCR)	4	<i>ampC, vanA, blaVIM-1, and ermB</i>	(Alexander et al. 2015)
Italy	Nardo	Real-time quantitative PCR (qPCR)	2	<i>tetO, and ermB</i>	(Böckelmann et al. 2009)
Portugal	Estarreja	16S rDNA	10	<i>blaTEM, tetA, tetB, strA, strB, sul1, sul2, dfrA1, aadA1 and aadA2</i>	(Araújo et al. 2017)
Romania	Cluj-Napoca	Real-time quantitative PCR (qPCR)	11	<i>aacC2, blaSHV, floR, ermA, mefA, sul1, sul2, tetA, tetC, tetO, and tetW</i>	(Szekeres et al. 2018)
Spain	Sabadell	Real-time quantitative PCR (qPCR)	5	<i>tetO, ampC, ermB, mecA, and blaSHV-5</i>	(Böckelmann et al. 2009)
United States	Urbana	Real-time quantitative PCR (qPCR)	7	<i>tetC, tetH, tetM, tetO, tetQ, tetW, and tetZ</i>	(Koike et al. 2007)
	Urbana	DGGE, 16S rRNA/rDNA	5	<i>tetM, tetO, tetQ, tetT and tetW</i>	(Chee-Sanford et al. 2001)
	Urbana	Real-Time quantitative PCR	4	<i>tetM, tetO, tetQ, and tetW</i>	(Mackie et al. 2006)

Abbreviations: PCR, polymerase chain reaction; DGGE, denaturing gradient gel electrophoresis

Table 2. The correlation reported among ARGs, antibiotics, microbial community, and physiochemical parameters in groundwater worldwide.

Country	Samples size (n)	Pollution source	Correlation variables	Correlation type	Remarks	Reference
(a) Antibiotics vs. ARGs						
China	20	Aquaculture	Sulfamethoxazole, ciprofloxacin, erythromycin, tetracycline vs. <i>sul2</i> , <i>tetG</i> , <i>tetX</i> , <i>tetQ</i> , <i>qnrB</i>	No association	Despite the low levels of antibiotics in groundwater, the relative abundance of ARGs in groundwater samples was even higher than surface waters. The effects of environmental parameters on ARGs were highly significant, reflecting the potential impact of these factors on the abundance of ARGs	(Tong et al. 2020)
China	5	Hospital waste, sewage treatment	Sulfamethoxazole, sulfamethazine, vs.	Negative	The highest relative abundance and highest frequency of sulfonamide resistance genes (<i>sul1</i>) were observed in groundwater followed by <i>sul2</i> , and	(Liu et al. 2019)

		plant	<i>sul1, sul2</i>		macrolide resistance genes (<i>ermB</i>).	
			Ciprofloxacin, norfloxacin, tetracycline vs. <i>ermB, tetC</i>	Positive		
China	10	Swine feedlots	Ciprofloxacin, norfloxacin vs. <i>qnrA</i>	Strong	The high concentrations of fluoroquinolones and tetracyclines in groundwater showed correlations with ARGs, with a strong correlation between fluoroquinolones resistance gene (<i>qnrA</i>).	(Li et al. 2018)
			Tetracycline, oxytetracycline, chlortetracycline vs. <i>tetG, tetM, tetO</i>	Significant		
(b) ARGs vs. environmental factors						
China	18	Swine farms	TP, TN, TOC, and NH ₄ -N vs. <i>sul2, tetG, qnrS</i>	Significant	The influence of ARGs on nutritional factors was observed to be more than physicochemical parameters.	(Gao et al. 2020)

China	20	Aquaculture	<i>tetM, tetC, ermB, ermK, sul2</i> vs. pH, DO, HCO_3^- , and DOC	Positive	The effects of environmental parameters on ARGs were highly significant, reflecting the potential impact of these factors on the abundance of ARGs.	(Tong et al. 2020)
China	10	Wastewater	<i>tetO, tetG, tetC, tetH, sul1, sul2, bla_{TEM}</i> vs. DO	Positive	ARGs in groundwater were significantly correlated with the water quality parameters such as DO, $\text{NH}_4\text{-N}$, and TOC, which may be greatly affected by human activities	(Wu et al. 2020)
			<i>tetO, tetG, tetC, tetH, sul1, sul2, bla_{TEM}</i> vs. $\text{NH}_4\text{-N}$, TOC, TN and TP	Negative		
(c) ARGs vs. microbial community						
China	18	Swine farms	<i>tetO, tetH, tetC, tetW, qnrS, sul1, sul2</i> vs. <i>Pseudomonas</i> &	Significant	The microbial diversity of groundwater was significantly decreased with the predominance of conditional pathogens <i>Acinetobacter</i> (up to 90%) in wells. The abundance of <i>Acinetobacter</i> was significantly correlated to bacterial	(Gao et al. 2020)

			<i>Acinetobacter</i>		abundance and ARGs.	
Romania	6	Sewage system, hospital wastewater	<i>bla</i> _{SHV} , <i>sul2</i> , vs. <i>Pedobacter</i> , <i>Acinetobacter</i> , <i>Pseudomonas</i> , and <i>Fluviicola</i>	Significant	The correlations between ARGs and specific taxa within the groundwater bacterial community were identified to be significant, revealing the potential hosts of resistance types.	(Szekeres et al. 2018)
China	5	Landfill leachate	<i>ampC</i> , <i>bla1</i> , <i>ermB</i> , <i>ermC</i> , <i>sul1</i> , <i>sul2</i> , <i>tetB</i> , <i>tetC</i> , <i>tetG</i> , <i>tetM</i> , <i>tetQ</i> vs. <i>Actinobacteria</i> , <i>Proteobacteria</i> , <i>Bacteroidetes</i>	Strong	Groundwater near municipal solid waste landfill is an underappreciated hotspot of antibiotic resistance and contribute to the ARGs spread via the flowing polluted groundwater	(Chen et al. 2017)
Germany	3	Wastewater treatment plant effluents	<i>blaVIM-1</i> , <i>vanA</i> , <i>ampC</i> , <i>ermB</i> , vs. <i>Enterococci</i> , <i>Staphylococci</i> , <i>P. aeruginosa</i> , & <i>Enterobacteriaceae</i>	No association	The abundances of most ARG targets were found to be increased in the microbial population after conventional wastewater treatment. As a consequence, some groundwater compartments displayed high abundances of all four ARGs.	(Alexander et al. 2015)

Canada	6	Manure application	<i>tetO</i> vs. <i>Campylobacter spp.</i>	Positive	Both <i>Campylobacter spp.</i> and <i>tetO</i> resistance genes were present in groundwater and their level increased after the application of liquid swine manure on clay loam.	(Frey et al. 2015)
Italy	5	Wastewater discharge	<i>tetO</i> , <i>ermB</i> , and <i>mecA</i> vs. <i>Helicobacter pylori</i> , <i>Yersinia enterocolitica</i>	Positive	The occurrence of <i>tetO</i> , <i>ermB</i> , and <i>mecA</i> in groundwater derived from artificial recharge suggests that groundwater may be a potential source of antibiotic resistance in the food chain.	(Böckelmann et al. 2009)
US	20	Swine feedlots	<i>tetT</i> , <i>tetO</i> vs. <i>Streptococcus spp.</i>	No association	Groundwater is a significant reservoir for the dissemination of resistance genes. It was found that <i>tet</i> genes occur in the environment as a direct result of agriculture and suggested that groundwater may be a potential source of antibiotic resistance in the food chain.	(Chee-Sanford et al. 2001)

Abbreviations: T, temperature; DO, dissolved oxygen; TP, total phosphorus; TN, total nitrogen; TOC, total organic carbon; NH₄-N, ammonia nitrogen; COD, chemical oxygen demand; EC, electrical conductivity

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